

Satyaki Tirumal

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EDUCATION

Mahindra University <i>B.Tech in Electronics and Computer Engineering</i>	Hyderabad, India Aug 2023 – Present
FIITJEE Jr. College, Miyapur <i>Intermediate (Class XII)</i>	Hyderabad, India Jun 2021 – May 2023

EXPERIENCE

TakeMe2Space Hackathon <i>Autonomous Satellite Control Developer – Lost in Space Track</i> - Built an autonomous Python imaging and attitude-control planner for Earth-observation satellites, using Inertial-Hold TSP and Mosaic Tiling scheduling to improve coverage and efficiency. - Modeled accurate satellite paths with SGP4 orbital propagation inside a Basilisk simulation.	2025
LAM Research Lab Hackathon <i>Semi-Finalist – All-India Level</i> Advanced to the semi-final round in a competitive, all-India level engineering hackathon.	2025
IoT & Robotics Training Programme <i>TechnooK × Cognizance'24, IIT Roorkee</i> Completed hands-on training in IoT sensor integration and robotics automation under IIT Roorkee's Cognizance programme.	Jul 2024 – Aug 2024

PROJECTS

UniShip <i>Founder & Team Lead Next.js, React, Flutter, Firebase, GroqAPI</i> - Founded and lead a student development team building a full-stack university placement platform spanning a Next.js web app and a Flutter mobile app, serving three distinct user roles (students, university admins, super admins). - Built a proctored online assessment portal with an in-browser code editor and a live invigilation dashboard, running a Dockerized code compiler on the college's HPC cluster. - Integrated the Groq API (Llama 3.3 70B) to auto-grade test submissions and power an AI resume builder with ATS scoring and PDF export. - Implemented three-tier role-based access control with Firebase Auth and Firestore, backed by real-time data sync.	2025 – Present
Flutter Task Manager <i>Flutter, Dart, Firebase Auth, Cloud Firestore, REST API</i> - Built a cross-platform task manager with full CRUD, real-time Firestore sync, and a motivational-quotes feed pulled from a third-party REST API. - Applied Material 3 design, form validation, and Firestore composite indexing to keep tasks sorted and the app responsive.	2026

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Dart, Python, Java, Kotlin, HTML, CSS
Frontend & Mobile: React, Next.js, Tailwind CSS, Flutter
Backend & Databases: Node.js, Express.js, Firebase (Auth, Firestore, Storage), MongoDB, REST APIs
Deployment & Tools: Docker, Vercel, Render, Railway, Git/GitHub, VS Code, Android Studio
AI/ML: LLM Integration & Prompt Engineering (Groq API), Retrieval-Augmented Generation (RAG), Machine Learning, Deep Learning, Python (Pandas, NumPy, scikit-learn)

ACTIVITIES

Represented Mahindra University in Football, Cricket, and Track & Field – competitive team play demanding coordination, discipline, and performance under pressure.
Volunteer, Blue Cross of Hyderabad – animal welfare community service.